

ELECTRONICS DEVICES (KEC-301)

A Project on AUDIO-RESPONSIVE LED SYSTEM

**Under the guidance of
DR. S.K. SRIWAS**

Submitted by

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**In partial fulfillment of the requirements for the award of the
degree of
BACHELOR OF TECHNOLOGY**



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

**BUNDELKHAND INSTITUTE OF ENGINEERING AND
TECHNOLOGY, JHANSI, INDIA 2022-2023**

CERTIFICATE

This is to certify that the project entitled “**AUDIO-RESPONSIVE LED SYSTEM**” has been carried out by **Prashant Kumar Gautam, Sandeep Sonkar, Shubham Pandey, Upendra Singh** under the supervision of Electronics and Communication Engineering at Bundelkhand Institute of Engineering and Technology, Jhansi. This project is part of the course **Electronics Devices** in the 2nd semester of Bachelor of Technology in Electronics and Communication Engineering.

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Acknowledgment

We would like to express our gratitude towards **Dr. S.K. Sriwas** for his guidance and constant supervision as well as for providing us necessary information regarding the project and this report. We feel thankful and express our kind gratitude towards our Director, Head of Department and all faculty members. We would also like to express our special gratitude and thanks to our parents for giving us constant support that improved our performance significantly.

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Dedication

“To My Beloved Parents and Almighty”

Declaration

We hereby proclaim that the work that is being presented in the project entitled “**AUDIO-RESPONSIVE LED SYSTEM**” is our work carried out by under the supervision of **Dr. S.K. Sriwas** of Electronics and communication Engineering at Bundelkhand Institute of Engineering and Technology, Jhansi. Information derived from the other sources has been quoted in the text and a list of references has been given.

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CONTENTS

<u>Content Details</u>	<u>Page No.</u>
Title page	1
Certificate	2
Acknowledgement	3
Dedication	4
Declaration	5
Contents	6
Introduction	7
Components Required	7
· BC547 Transistor	8
· Battery	8
· Miceroophone	9
· Switch	9
· Led	10
· Capacitor	11
· Resistor	11
Circuit Diagram	12
Working	12
Implementation	13
Applications	13
Conclusion	14
References	14

AUDIO-RESPONSIVE LED SYSTEM

- In this, we are going to show you how to make Music Rhythm Operated Dancing Light using LEDs & Transistors. You might have seen the Disco Lights or DJ lights or light during a function that turns ON and OFF according to the beats or Rhythm of the music.
- This Music Rhythm Operated Dancing Light circuit is based on transistor BC547. This circuit is very simple and easy to build, it just requires a few basic components.
- These lights turn ON & OFF according to the length and pitch of music beats or rhythm. Basically these are designed to pick the high-intensity sound like Bass sound. So these lights follow the high pitch beats in music like drum beats and Turn ON and OFF according to music pattern.

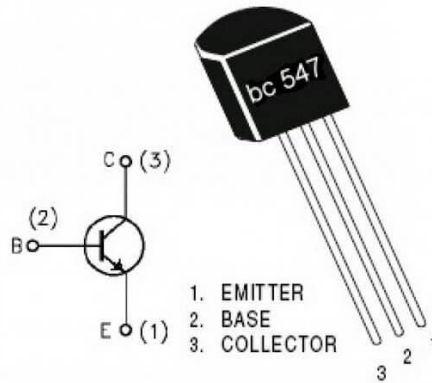
COMPONENTS REQUIRED:

The circuit requires a few of the basic components which are listed below:

1. Transistors BC547
2. LED1
3. Capacitor - 100nF or 0.1uF
4. Resistor 1M
5. Resistor 10K
6. Resistor 22 Ω
7. Microphone
8. Switch ON/OFF
9. 9 Volt DC Power Supply

I. BC547 TRANSISTOR

- BC547 is a normal NPN (Negative-Positive-Negative) junction transistor.
- It is a BJT transistor and is often used to satisfy the need of quick switching.
- BC547 is usually used for current amplifier, quick switching and pulse-width modulation (PWM).



II. BATTERY

- An electrical battery is a combination of one or more electrochemical cells, used to convert stored chemical energy into electrical energy.
- The battery has become a common power source for many household, robotics and industrial applications.
- Larger batteries provide standby power for telephone exchanges or computer data centers.



II. MICROPHONE:

- The microphone is a transducer device which converts sound energy into electrical energy. Microphones are often referred to a **MIC**.
- A microphone is used to capture some sort of sound and produce an electrical signal according to it.



III. SWITCH:

- A switch responds to an external force to mechanically change an electric signal.
- Switches are used to turn electric circuits ON and OFF and to switch electric circuits.



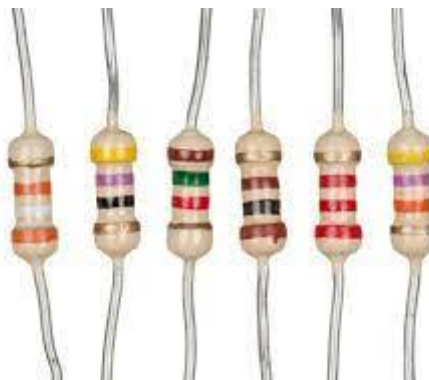
IV. LED:

- LED, in full light-emitting diode, in electronics, a semiconductor device that emits infrared or visible light when charged with an electric current.
- Visible LEDs are used in many electronic devices as indicator lamps, in automobiles as rear-window and brake lights, and on billboards and signs as alphanumeric displays or even full-colour posters.
- Infrared LEDs are employed in autofocus cameras and television remote controls and also as light sources in fibre-optic telecommunication systems.



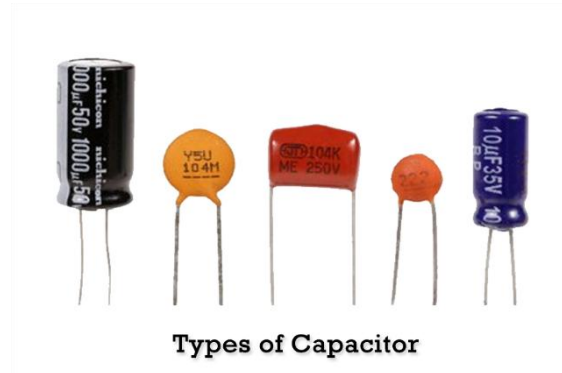
V. RESISTOR:

- A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element.
- In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other



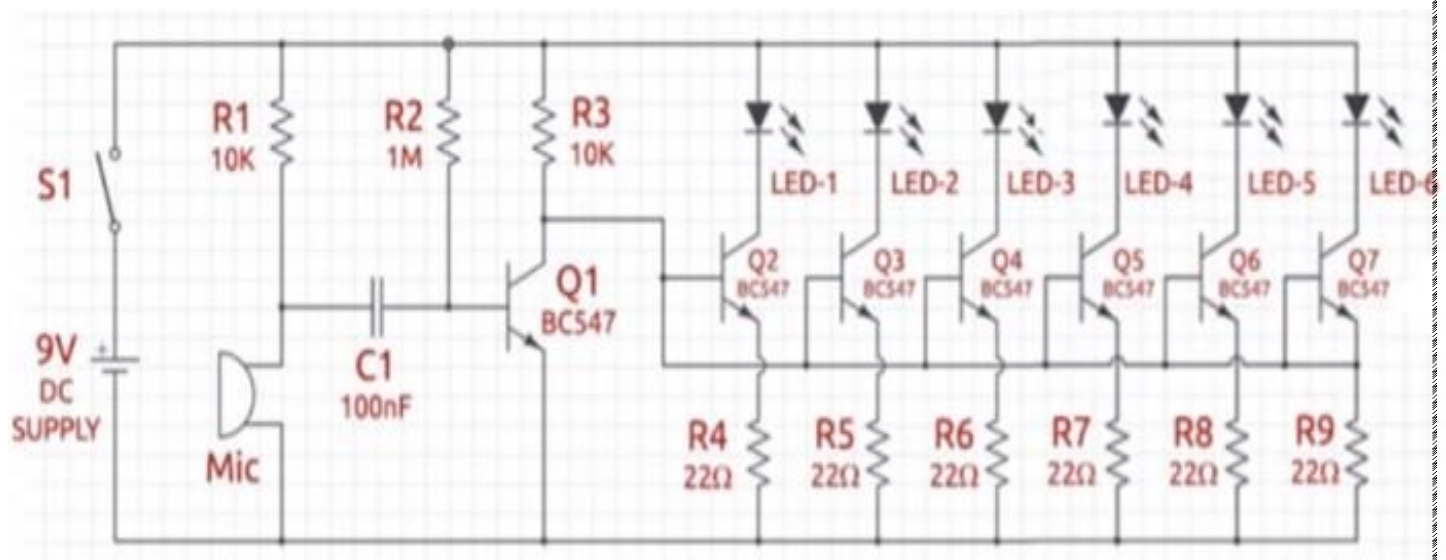
VI. CAPACITOR

- A capacitor is a passive two-terminal electrical component that stores potential energy in an electric field. The effect of a capacitor is known as capacitance.
- It is commonly used for removing noise or making the supply voltage of a circuit more stable.



CIRCUIT DIAGRAM:

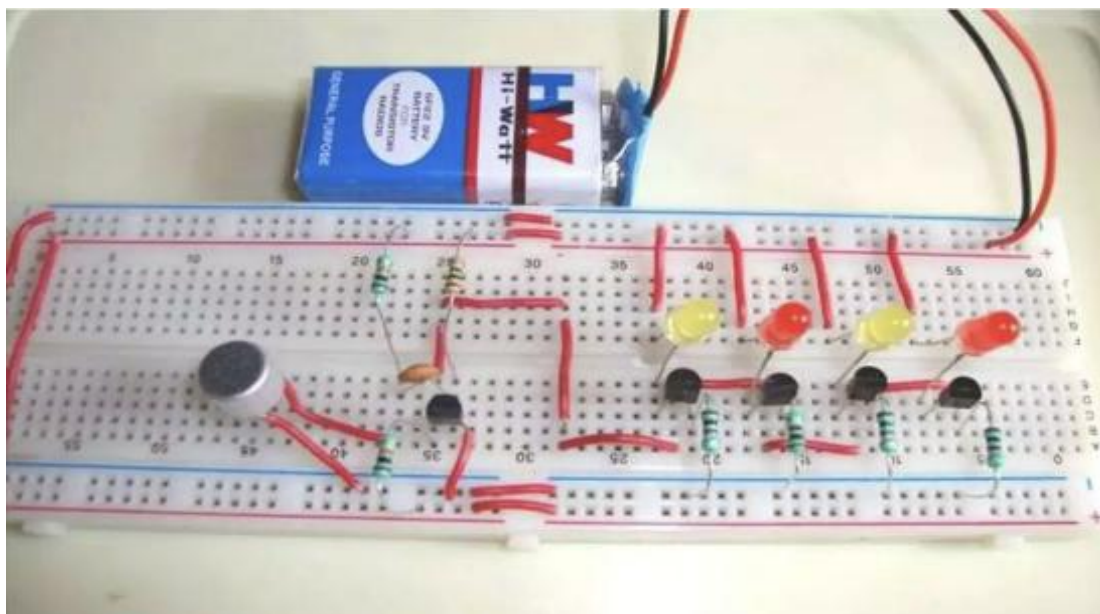
Assemble the circuit on the breadboard as per the circuit diagram.



WORKING OF MUSIC RHYTHM OPERATED DANCING LIGHT:

- In this Simple Music Rhythm Operated Dancing Light, mic. picks up the sound signals & converts them into a voltage signal. The voltage developed is directly proportional to the quantity of sound falling on the mic.
- These voltage signals are further fed into High Pass RC Filter R1 & C1 for eliminating the noise from the sound. Further, an NPN transistor Q1 BC547 works as a signal amplifier.
- Then finally these music signals are given to the array of six transistors grounded across with 22 Ω resistor.
- The transistor in this array works as an amplifier, and glows the four LEDs according to the sound pattern. We can also add more LEDs & transistors to make it following more longer pattern.

IMPLEMENTATION:

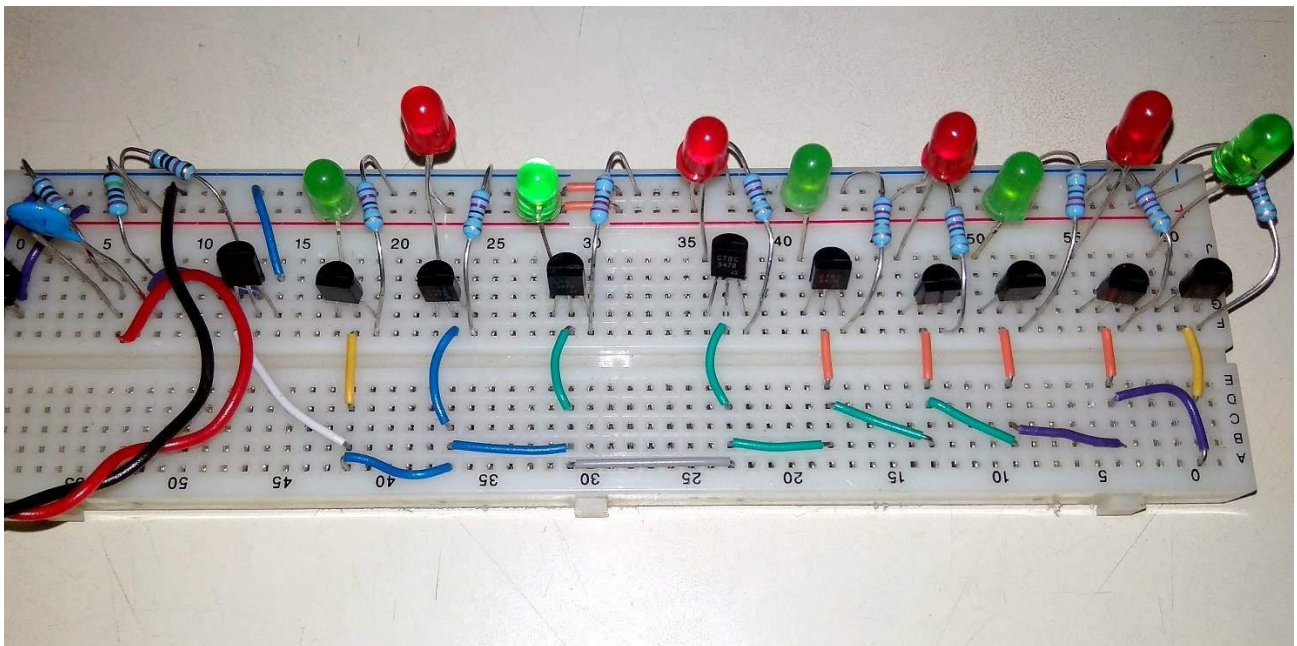


APPLICATIONS:

- Dancing LED circuit can be used for any visual sign indication in any highways or it can be used in advertisement boarding also.
- LED blinking circuit can be used in signaling purpose (It can be used as signal for help, if you are in danger)
- LED blinking circuit can be used as flashing beacon.
- There are lots of applications with this circuit.
- It's used for celebration like wedding anniversary, birthday, festivals.

RESULT:

Our result



CONCLUSION:

- This circuit has been designed as a prototype.
- It can be used for large-scale projects and commercial/domestic use.
- This circuit works on DC volt.
- We have conclude that this circuit works on low pass and high pass filter phenomenon and had come to know that how the LED Dancing Light works and what are the structure of it.
- We can used this circuit like disco lights.

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